TALASH

Talent Acquisition & Learning Automation for Smart Hiring

Milestone 2 Report - Core Extraction, Analysis Pipeline & Intermediate Web App

Course: CS 417 - Large Language Models (Spring 2026)
Program: BSDS-2023 | SEECS, NUST
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1. Introduction

Recruitment is one of the most critical processes in any organization, yet it remains time-consuming, subjective, and heavily dependent on manual evaluation. Traditional hiring workflows rely on basic keyword matching and human judgment, often overlooking deeper aspects of a candidate's academic quality, research output, publication profile, experience consistency, and skill relevance.

TALASH (Talent Acquisition & Learning Automation for Smart Hiring) is an AI-powered recruitment support system focused on university hiring needs, built using Large Language Models (LLMs). The system automates CV screening, candidate-job matching, academic and publication analysis, experience validation, and skill alignment assessment - helping recruiters make more informed, evidence-based, and efficient hiring decisions.

This report documents Milestone 2 of the TALASH project. Building on the preprocessing foundation of Milestone 1, Milestone 2 delivers a fully functional analysis pipeline covering educational profile analysis, professional experience analysis, research profile processing, missing information detection with personalized email drafting, initial candidate summary generation, and an intermediate Streamlit web application with tabular outputs, interactive charts, and a candidate comparison dashboard.

2. Milestone 2 Objectives

The specific goals for Milestone 2, as defined in the project rubric, are:

- Implement a working CV ingestion pipeline with folder-based batch reading
- Enhance CV parsing with LLM-assisted structured extraction (Google Gemini)
- Functional implementation of Educational Profile Analysis with CGPA normalization, gap detection, institution quality lookup, and progression scoring
- Functional implementation of Professional Experience and Employment History analysis with timeline normalization, overlap detection, gap detection with severity classification, and career progression assessment
- Missing information detection with three-tier classification (missing, incomplete, unclear) and personalized email drafting per candidate
- Initial candidate summary generation with suitability label
- Partial Research Profile processing - publication counts, journal vs. conference classification, and topic analysis
- Tabular outputs for education, experience, and publications
- Initial charts and graphs - score charts, publication timelines, comparison radar charts
- Intermediate Streamlit web application with 7-tab interface and candidate selector

3. System Architecture - Milestone 2 Updates

The Milestone 1 architecture has been significantly expanded. The Analysis Layer now contains five active modules, and the Presentation Layer has been rebuilt into a 7-tab interactive dashboard with LLM-generated summaries, Plotly charts, and multi-candidate comparison.

3.1 Updated Architecture Overview

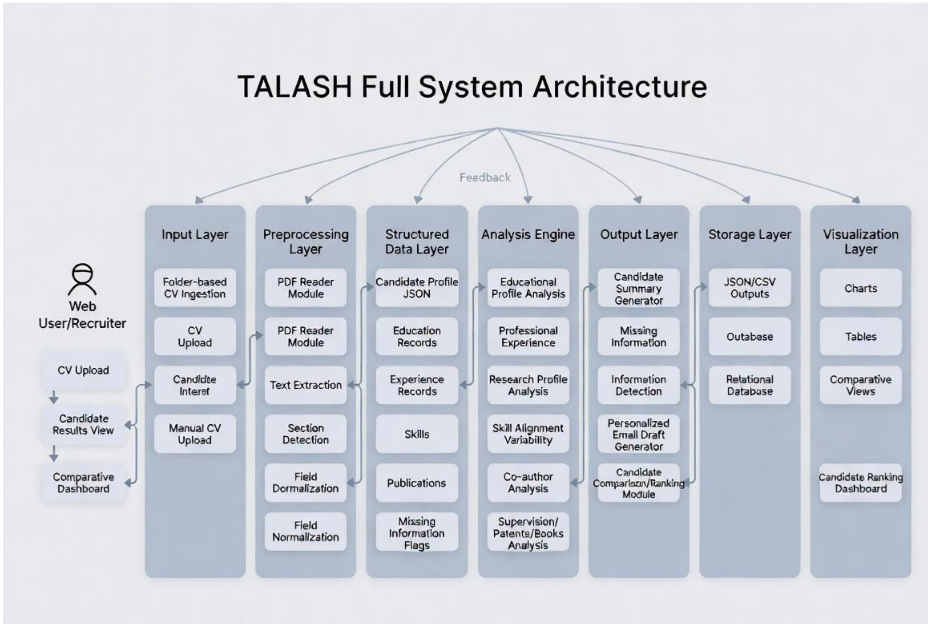


Figure 1: TALASH System Architecture (updated for Milestone 2)

3.2 New Module Descriptions

analysis/education_analysis.py

Full educational profile analysis module. Takes the parsed profile dict and performs: CGPA/percentage/grade normalization across different scales, education level classification (SSC → HSSC → UG → PG → PhD), gap detection between education levels with justification via work experience, academic progression trend analysis, specialization consistency checking, institution quality lookup using a curated tier database of Pakistani and international universities, and overall academic strength assessment. Returns a structured dict used by the Streamlit education tab.

analysis/experience_analysis.py

Professional experience analysis module. Normalizes employment dates from duration strings (e.g., 'Feb-2014 - Sep-2024') with proper handling of 'Present' end dates, builds a sorted employment timeline, detects job-job overlaps with severity classification, detects employment gaps (> 90 days) with minor/moderate/significant severity labels, cross-checks gaps against education records for justification, performs career progression scoring using a role-seniority lookup, and generates a human-readable continuity summary paragraph.

analysis/email_drafter.py

Missing information detection and personalized email drafting module. Classifies all parsed profile fields into three tiers - missing (field entirely absent), incomplete (field exists but lacks sub-fields), and unclear (value is ambiguous). Generates a candidate-specific follow-up email with subject and body, listing only the sections that have issues. A batch function handles multiple candidates and returns individualized emails for each.

analysis/research_analysis.py

Partial research profile processing module (Milestone 2 scope). Counts total publications, classifies by type (journal vs. conference), extracts impact factors, identifies top publication venues, builds a year-wise publication timeline, and identifies dominant research themes by keyword analysis across titles.

analysis/summary_generator.py

Candidate summary generation module. Synthesizes outputs from all analysis modules into a concise profile summary, assigns a suitability label (Strong Candidate, Good Candidate, Needs Review, Insufficient Data) based on academic strength, experience years, and research output, and generates an LLM-enhanced narrative summary using Google Gemini when available.

analysis/normalizers.py

Shared normalization utilities used across analysis modules. Handles CGPA scale detection and normalization, education level classification, specialization extraction from degree strings, and safe integer parsing.

llm/gemini_client.py

Google Gemini API client. Provides LLM-enhanced parsing with structured JSON output extraction. Used as the primary parser in Milestone 2 with regex-based parser as fallback when the API is unavailable.

4. CV Parsing and Structured Extraction - Enhancements

The Milestone 1 regex-based parser has been significantly enhanced in Milestone 2 with LLM-assisted extraction using Google Gemini. The parser now operates in two modes: LLM-first with regex fallback.

4.1 LLM-Enhanced Parsing

When the Gemini API is available, the parser sends the full extracted CV text to the model with a structured prompt requesting a JSON response covering personal information, education, experience, publications, skills, supervision records, books, and patents. The response is parsed and validated against the expected schema. This significantly improves extraction accuracy for free-form academic CVs that do not follow the NUST HR portal table format.

4.2 Structured Extraction Schema - Milestone 2 Extensions

Field	Type	Description
personal_information	Object	Name, email, phone, DOB, position applied for
education	Array	Degree, grade, year, institution, specialization
experience	Array	Title, org, duration, start/end year
publications	Array	Title, venue, type, year, impact factor, authorship
skills	Array	Technical and domain skills
supervision	Array	MS/PhD students supervised (new in M2)
books	Array	Authored/co-authored books with ISBN (new in M2)

Field	Type	Description
patents	Array	Patent numbers, dates, countries (new in M2)

Table 1: Extended JSON Schema - Milestone 2

4.3 Folder-Based Batch Processing

The system now supports processing all CVs in the data/input_cvs/ folder in a single operation. A progress bar tracks processing status and results are stored in session state for cross-candidate comparison. The sidebar allows switching between processed candidates without re-running the pipeline.

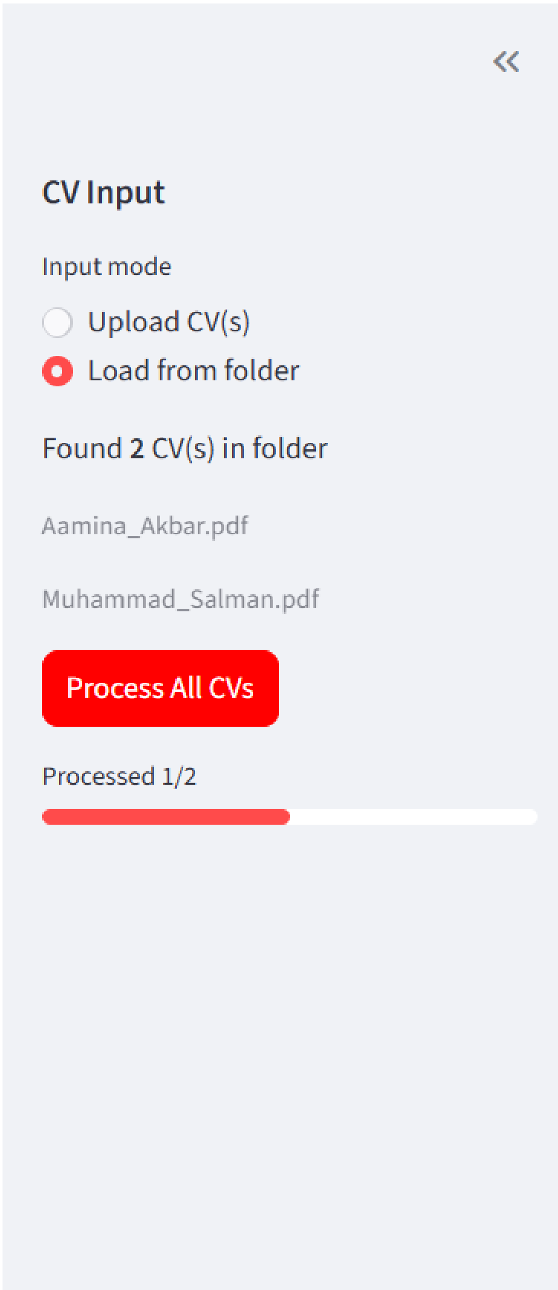


Figure 2: Folder-based CV processing with progress indicator

5. Educational Profile Analysis

The educational profile analysis module implements a comprehensive evidence-based assessment of each candidate's academic background, covering score normalization, level classification, gap detection, institution quality, and overall strength interpretation.

5.1 Score Normalization

Candidates report academic results in different forms - CGPA on 4.0 or 5.0 scales, percentages, grades, or divisions. The normalizer module converts all representations to a common scale of 0–100 for fair comparison:

- CGPA on 4.0 scale → multiplied by 25
- CGPA on 5.0 scale → multiplied by 20
- Percentages → used directly
- Letter grades (A, B+, etc.) → mapped to midpoint percentages
- Original values are preserved alongside normalized values

5.2 Education Level Classification and Gap Detection

Each education record is classified into one of five levels: SSC, HSSC, UG, PG, or PhD. The system detects gaps between consecutive education levels and compares them against expected durations (e.g., HSSC to UG should be approximately 4 years). Gaps beyond expected duration by more than one year are flagged. The module then cross-references flagged gaps against the candidate's work experience to determine justification.

5.3 Institution Quality Assessment

Institutions are assessed against a curated tier database covering Pakistani and international universities. The database assigns a tier (1 = Highly Ranked, 2 = Well Ranked, 3 = Recognized, 4 = Regional) and an approximate QS/THE ranking range where available. This enables recruiters to quickly assess the calibre of institutions attended without manually looking up rankings.

5.4 Education Tab - Screenshots

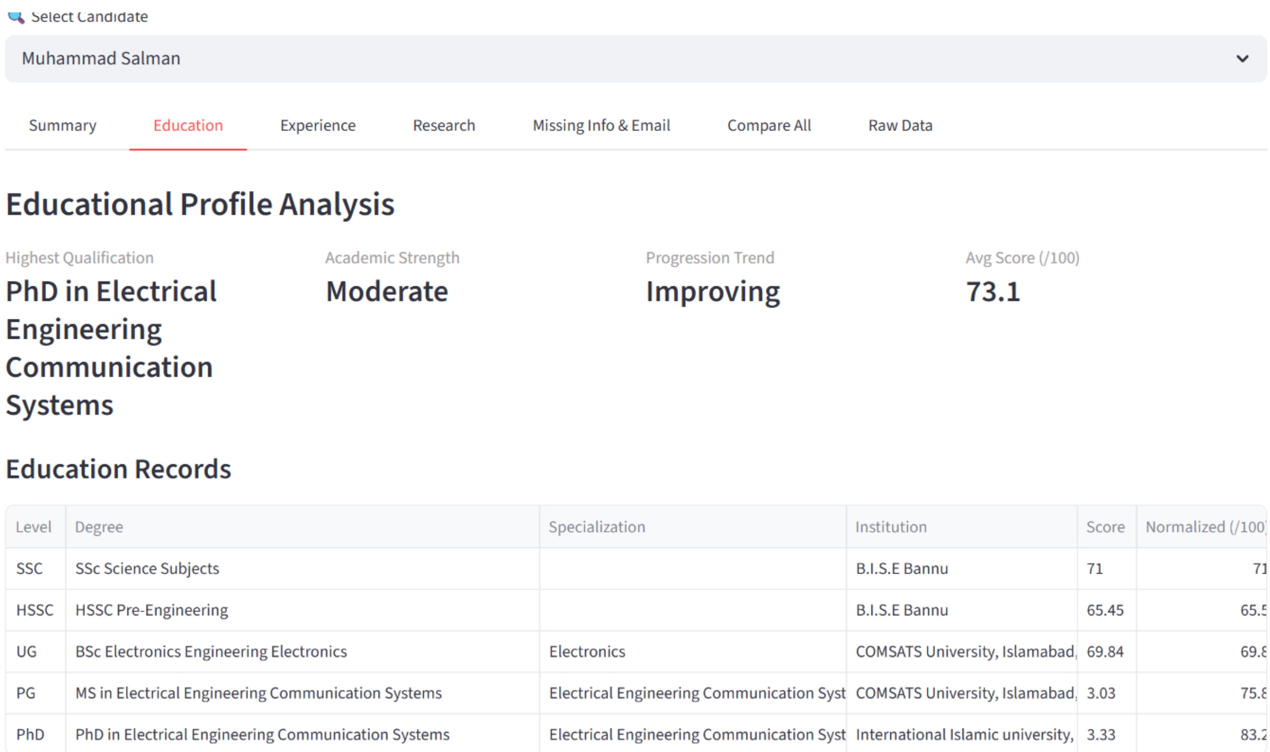


Figure 3: Education Tab - normalized scores table, degree sequence, institution quality, and gap analysis

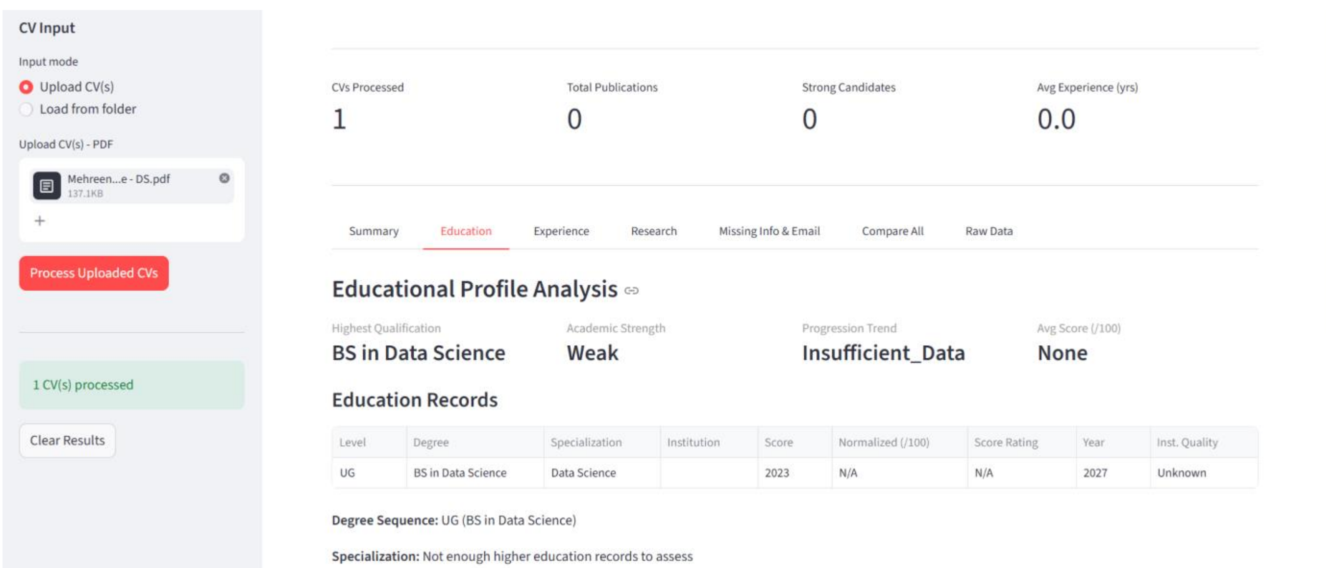


Figure 4: Education analysis for uploaded CV - metrics cards showing highest qualification, strength, and trend

6. Professional Experience and Employment History Analysis

The experience analysis module evaluates the candidate's professional profile, employment continuity, career progression, and timeline consistency.

6.1 Timeline Normalization

All experience entries are normalized from multiple date formats (e.g., 'Feb-2014 - Sep-2024', year-only, 'Present') into comparable datetime objects. Entries are sorted chronologically to build an ordered employment timeline.

6.2 Gap and Overlap Detection

The module scans the ordered timeline for two types of issues:

- Employment gaps - periods of more than 90 days between consecutive roles. Each gap is classified as minor (3–12 months), moderate (12–24 months), or significant (> 24 months). Gaps are checked against education records for justification.
- Job-job overlaps - pairs of roles whose date ranges intersect. Overlaps of 3 months or less are flagged as minor transition overlaps (likely acceptable); longer overlaps are flagged as concurrent roles requiring clarification.

6.3 Career Progression Scoring

Each job title is matched against a seniority lookup table (Intern=1 through Dean=7). The module compares average seniority scores in the first half vs. second half of the career to classify progression as: strong upward, moderate upward, lateral, or downward movement.

6.4 Experience Tab - Screenshots

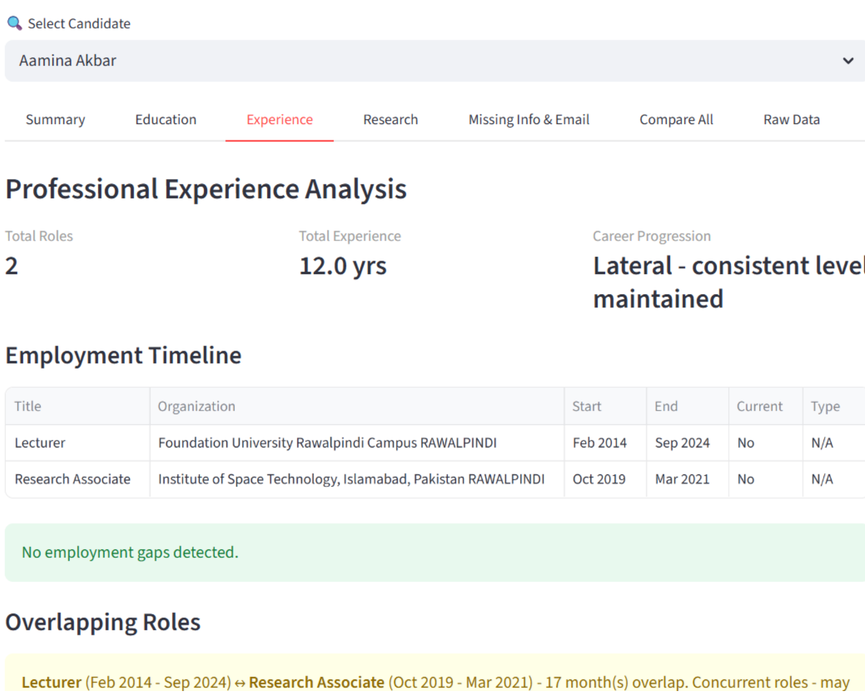


Figure 5: Experience Tab - ordered timeline table, gap detection, overlap analysis, and career progression label

7. Research Profile Analysis (Partial - Milestone 2 Scope)

Milestone 2 includes partial research profile processing. The research analysis module processes each candidate's publication list to provide initial quantitative insights.

- Total publication count with breakdown by type (Journal, Conference, Other)
- Year-wise publication timeline showing research output over time
- Impact factor extraction and average impact factor calculation for journal papers
- Dominant research theme identification by keyword frequency analysis across publication titles
- Top publication venues listed by frequency

Full research profile analysis - including WoS/Scopus indexing verification, quartile ranking, authorship role analysis (first author vs. corresponding author vs. co-author), conference A* ranking, co-author network analysis, and topic variability scoring - is planned for Milestone 3.

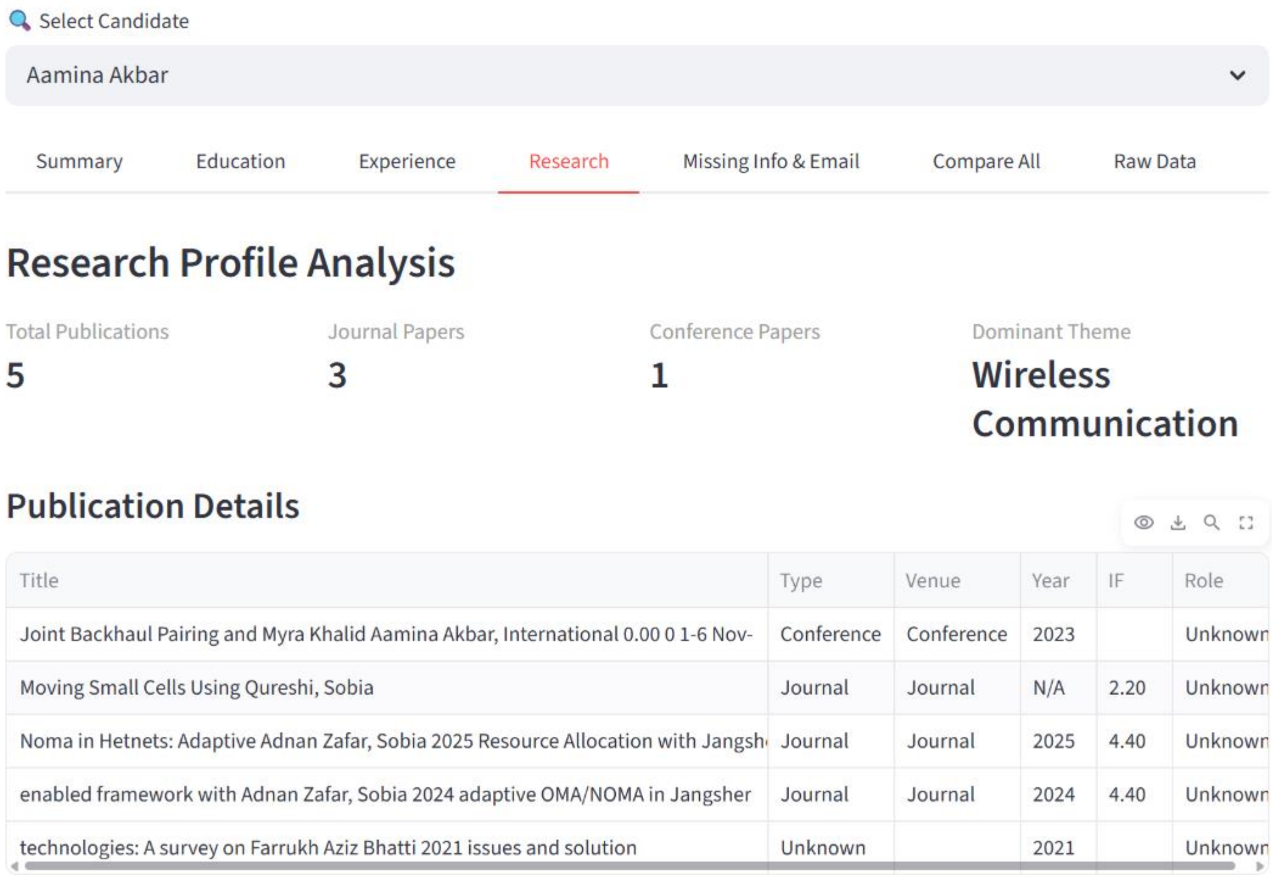


Figure 6: Research Tab - publication counts, type breakdown, impact factors, and dominant theme

8. Missing Information Detection and Personalized Email Drafting

The missing information module implements a three-tier classification system and generates individualized follow-up emails for each candidate.

8.1 Three-Tier Classification

Tier	Label	Description
1	Missing	Field is entirely absent from the CV - email, phone, DOB, education, experience
2	Incomplete	Field exists but lacks key sub-fields - grade missing for a degree, end date missing for a role
3	Unclear	Value exists but is ambiguous - publication year embedded in title, grade format unreadable

Table 2: Three-Tier Missing Information Classification

8.2 Personalized Email Generation

For each candidate with actionable missing items, the system generates a personalized email with a specific subject line and structured body. The email only lists the sections that have issues - candidates with no missing information receive no email. Each email is addressed by the candidate's name extracted from the CV.

8.3 Missing Info Tab - Screenshot

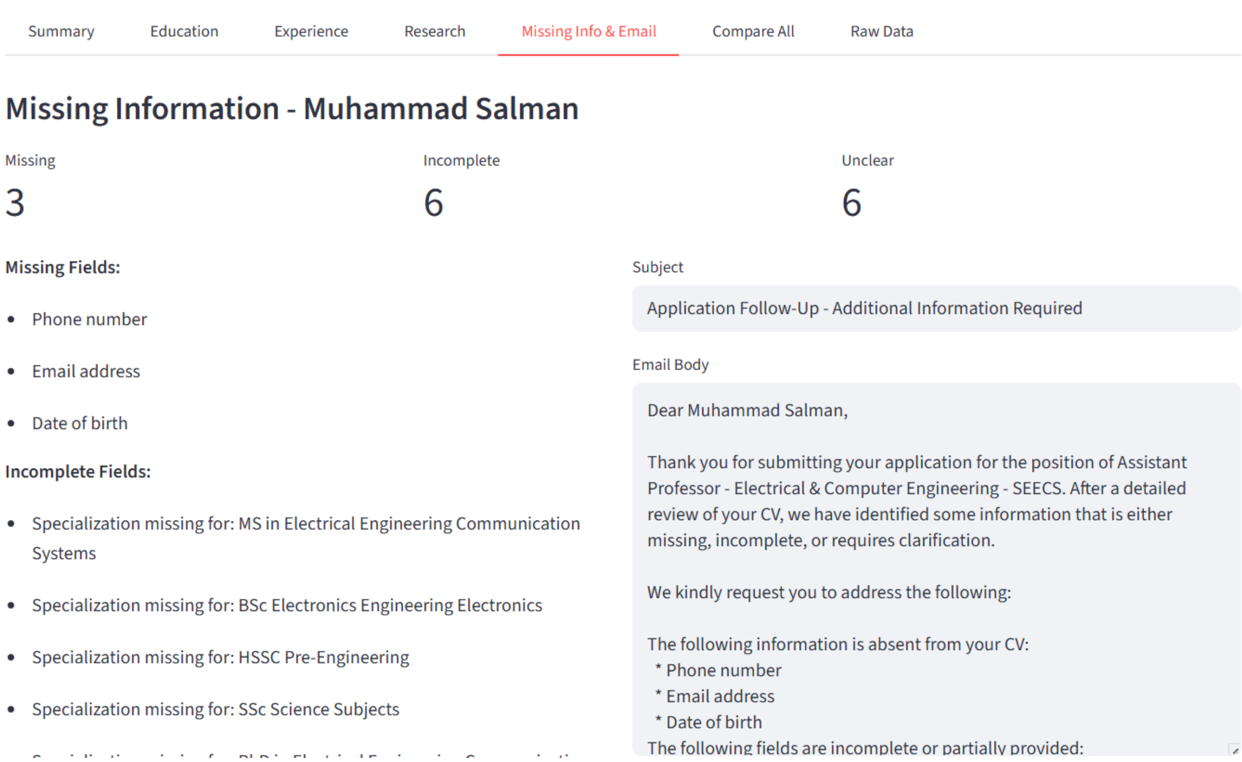


Figure 7: Missing Info & Email Tab - classified issues and generated personalized email draft

9. Candidate Summary Generation

The summary generator synthesizes outputs from all analysis modules into a concise candidate profile summary. It assigns a suitability label based on three weighted factors:

- Academic strength (strong / moderate / fair / weak) from the education analysis
- Total years of professional experience from the experience analysis
- Research output (total publications) from the research analysis

Suitability labels assigned are: Strong Candidate, Good Candidate, Needs Review, or Insufficient Data. An LLM-generated narrative summary provides a human-readable interpretation of the full profile. This summary is displayed in the Summary tab and used in the comparison dashboard.

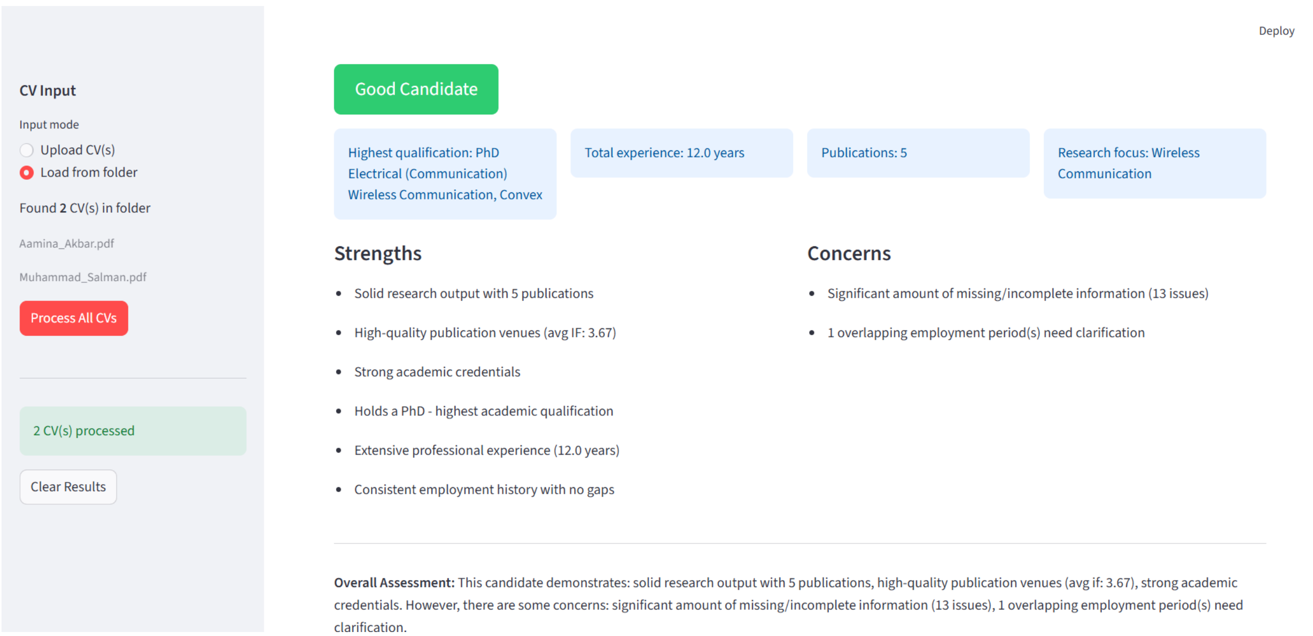


Figure 8: Summary Tab - suitability label, metric cards, and LLM-generated narrative summary

10. Intermediate Web Application

The Streamlit web application has been completely rebuilt for Milestone 2 with a 7-tab interface, dual input modes, an analytics dashboard with metric cards, Plotly interactive charts, and a multi-candidate comparison view.

10.1 Application Home and Input Modes

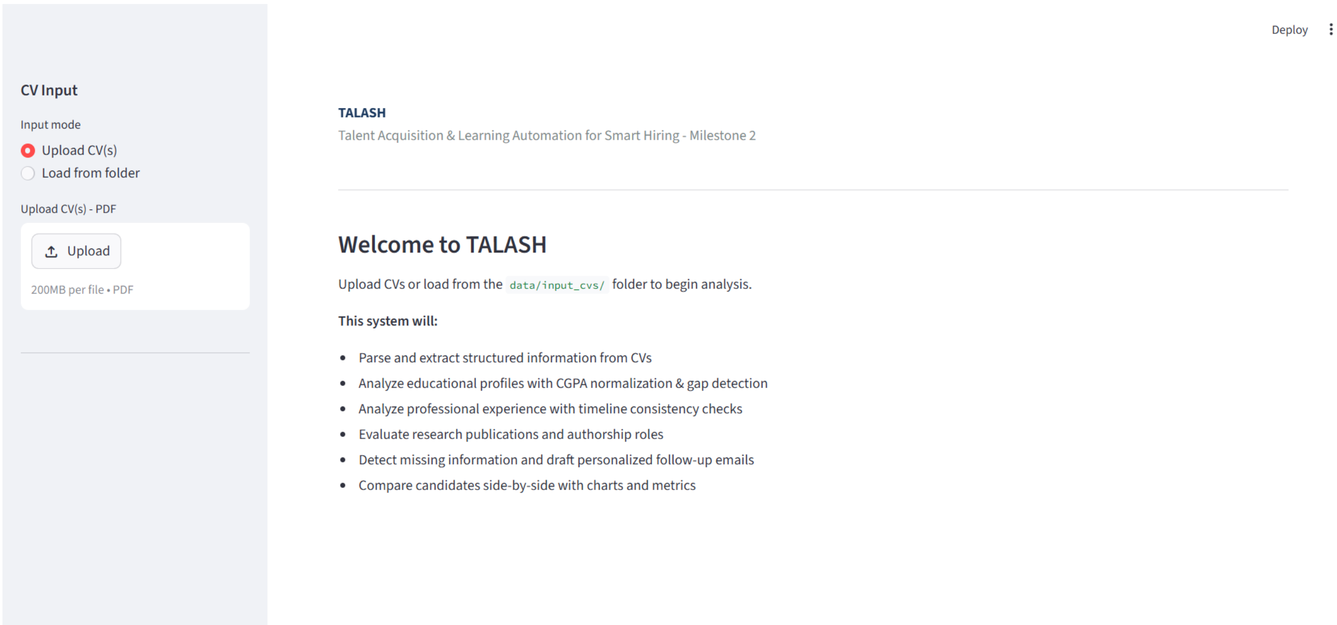


Figure 9: TALASH Home Screen - upload and folder input modes with feature overview

10.2 Dashboard Metrics Cards

After processing, the dashboard shows four summary metric cards: total CVs processed, total publications across all candidates, count of Strong Candidates, and average experience in years. These cards provide an at-a-glance overview before drilling into individual candidate tabs.

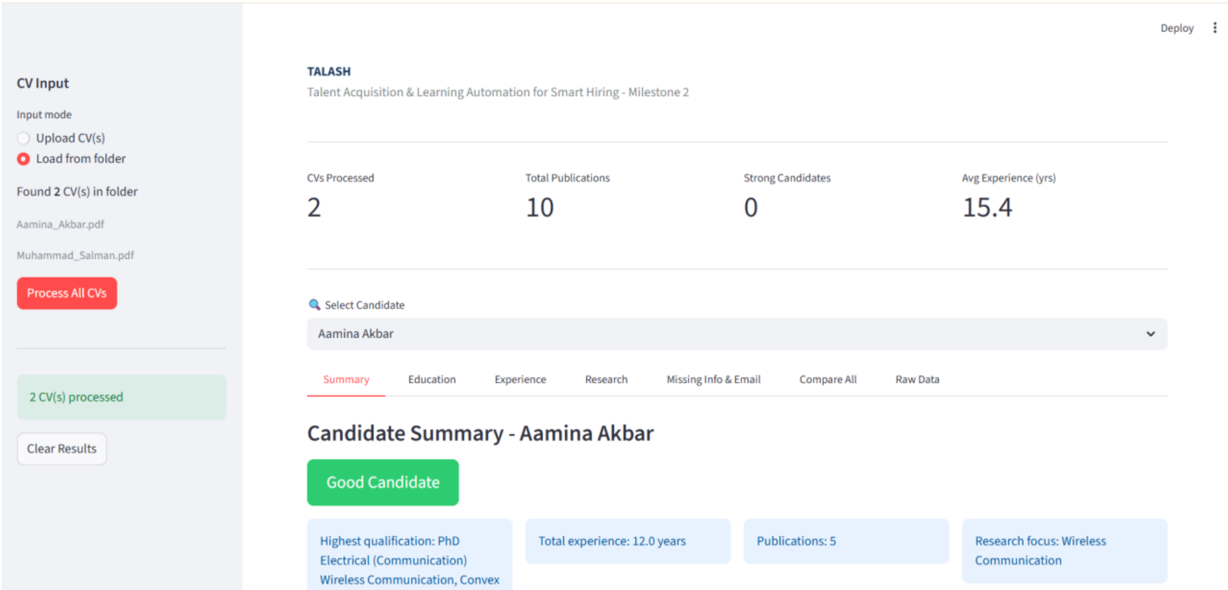


Figure 10: Dashboard metric cards - aggregate statistics across all processed CVs

10.3 Upload Mode

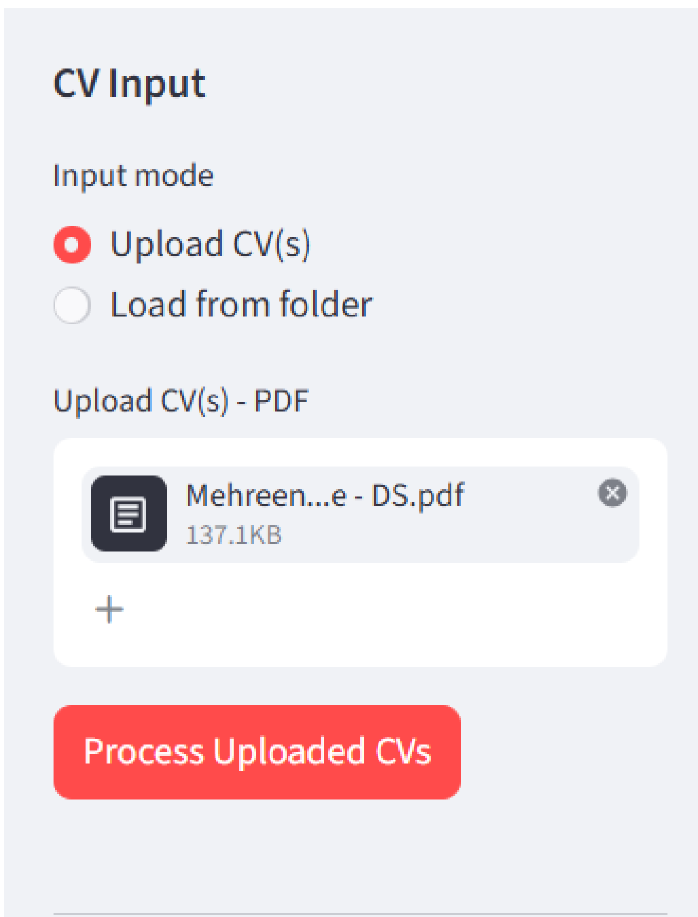


Figure 11: CV upload widget - accepts multiple PDFs simultaneously

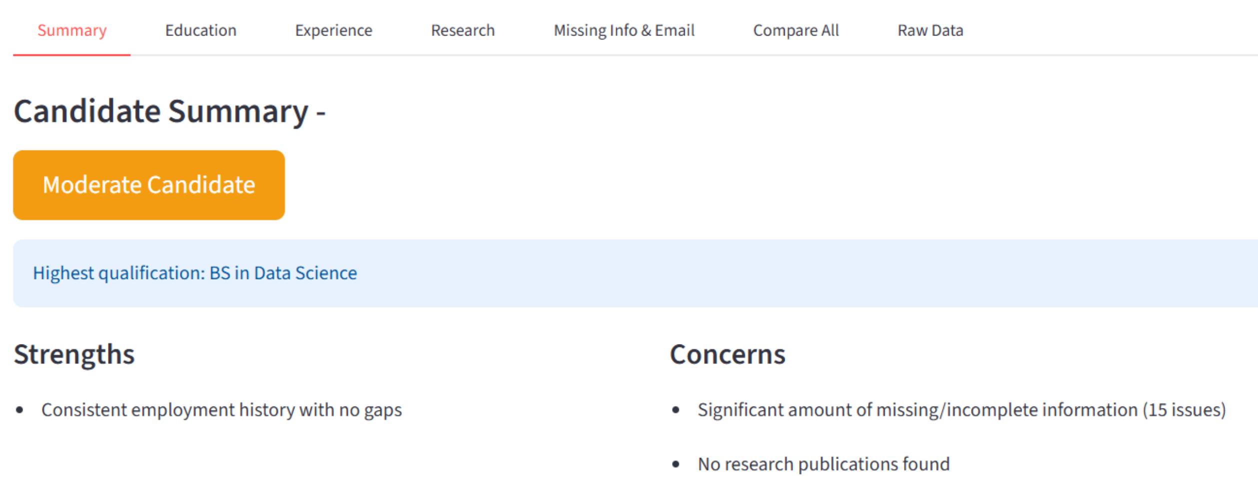


Figure 12: Post-upload dashboard showing processed candidate metrics

10.4 Candidate Selector

When multiple CVs have been processed (via upload or folder mode), a candidate selector appears allowing the recruiter to switch between candidates and view their full analysis across all tabs without re-processing.

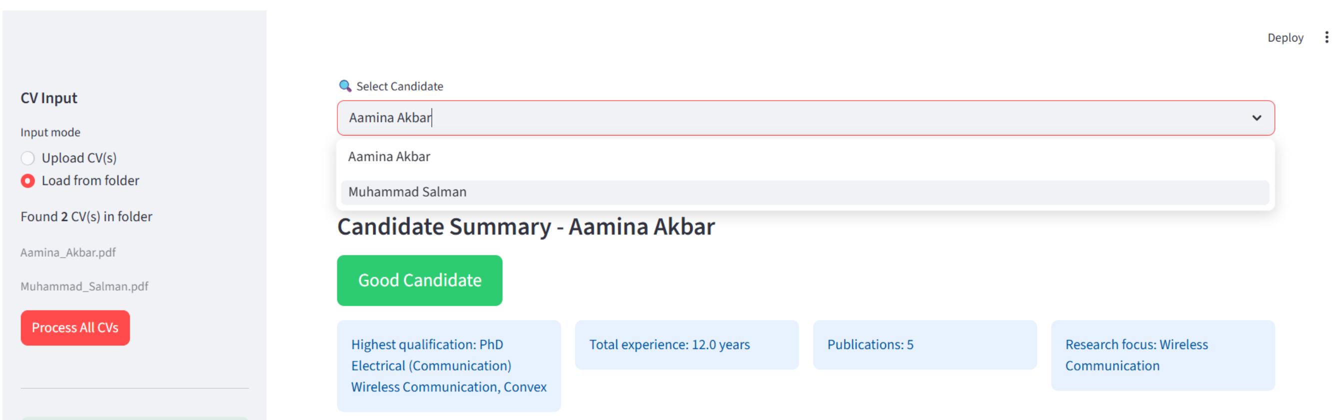


Figure 13: Candidate selector - switching between processed candidates

10.5 Raw Data Tab

It shows the data parsed data in json format.

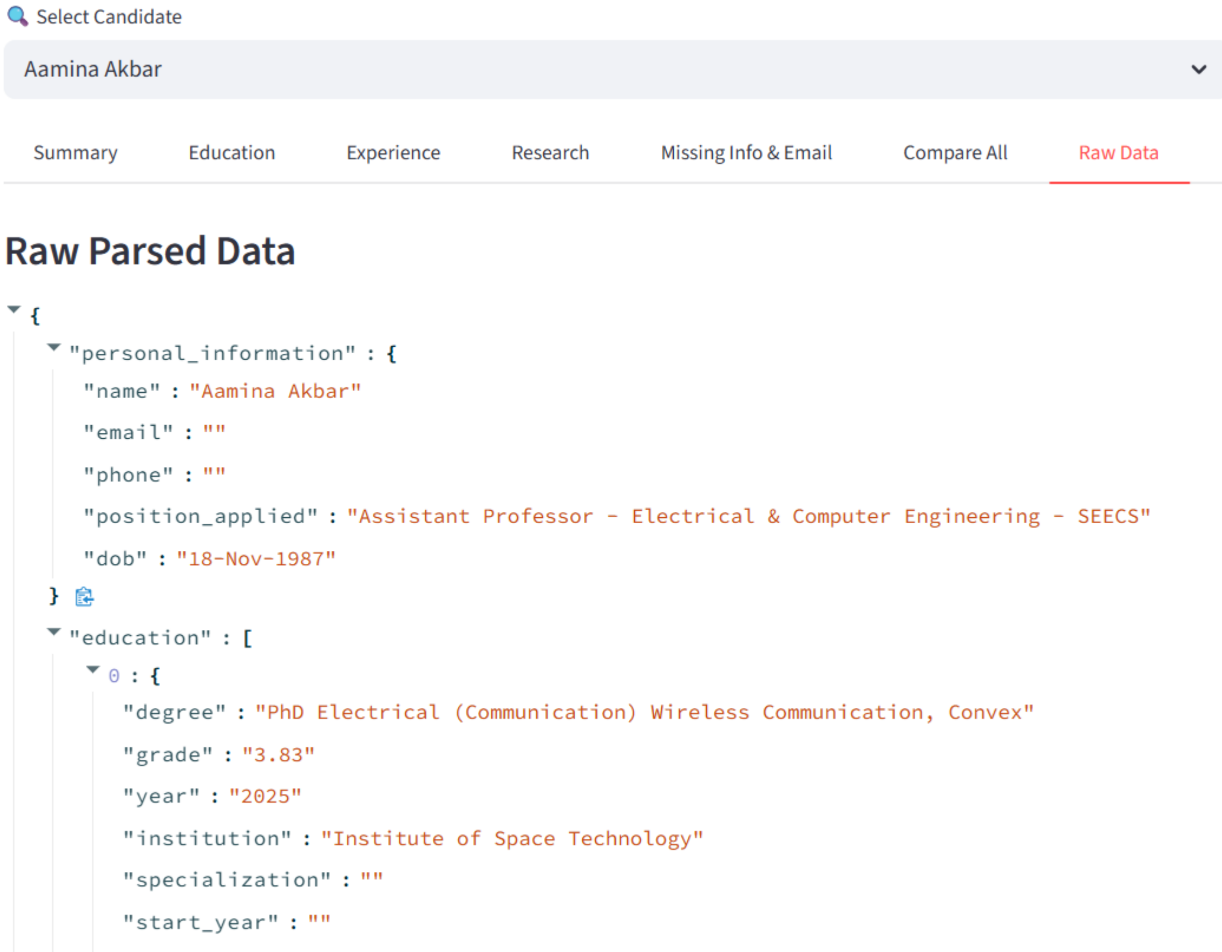
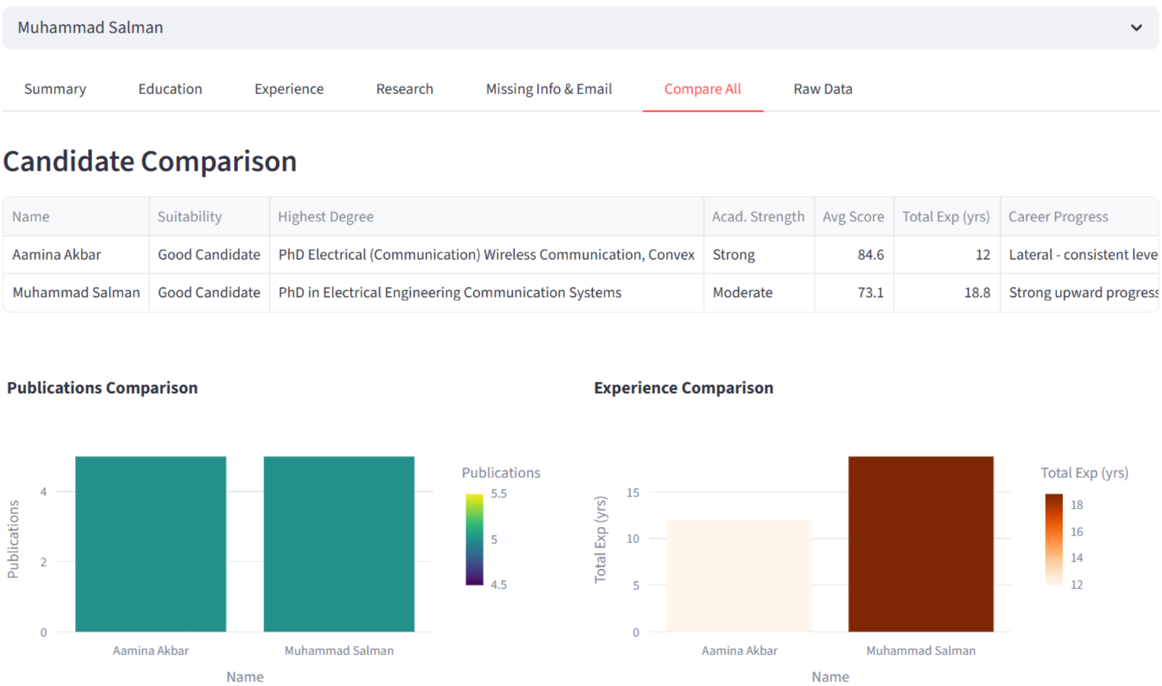


Figure 14: Raw Data Tab - full parsed JSON profile for inspection and debugging

10.6 Comparison Tab

The Compare All tab renders a side-by-side comparison table for all processed candidates covering highest degree, academic strength, total experience, publication count, and suitability label. A Plotly radar chart provides a visual multi-dimensional comparison.



Name	Suitability	Highest Degree	Acad. Strength	Avg Score	Total Exp (yrs)	Career Progress
Aamina Akbar	Good Candidate	PhD Electrical (Communication) Wireless Communication, Convex	Strong	84.6	12	Lateral - consistent leve
Muhammad Salman	Good Candidate	PhD in Electrical Engineering Communication Systems	Moderate	73.1	18.8	Strong upward progress

Figure 15: Compare All Tab - candidate comparison table and radar chart

11. Technologies Used

Technology	Version	Purpose
Python	3.14	Primary programming language
Streamlit	≥ 1.33.0	Web application framework and 7-tab UI
Google Gemini API	gemini-2.0-flash	LLM-enhanced CV parsing and summary generation
pdfplumber	≥ 0.11.0	Layout-aware PDF text extraction
Plotly	Latest	Interactive charts - bar, pie, radar
pandas	≥ 2.2.0	Tabular data handling and display
python-dateutil	≥ 2.9.0	Date parsing for experience timelines
re (stdlib)	Built-in	Regex-based field extraction fallback
json (stdlib)	Built-in	JSON serialization and LLM response parsing
Git / GitHub	-	Version control and team collaboration

Table 3: Technologies and Libraries - Milestone 2

12. Repository Structure

Path	Description
app/streamlit_app.py	7-tab Streamlit web application (rebuilt M2)
preprocessing/pdf_reader.py	PDF text extraction using pdfplumber
preprocessing/parser.py	LLM + regex hybrid CV parsing pipeline
analysis/education_analysis.py	Educational profile analysis (Member 1)
analysis/experience_analysis.py	Experience and employment analysis (Member 2)
analysis/email_drafter.py	Missing info detection and email drafting (Member 2)
analysis/research_analysis.py	Research profile processing (Member 3)
analysis/summary_generator.py	Candidate summary and suitability scoring (Member 3)
analysis/normalizers.py	Shared normalization utilities (Member 1)
llm/gemini_client.py	Google Gemini API client
data/input_cvs/	Sample CV PDFs for folder-based batch processing
data/sample_outputs/	Pre-parsed JSON profiles
data/analysis_outputs/	Analysis results JSON files
docs/	Architecture diagrams and wireframes
requirements.txt	Python dependency specifications

Table 4: Repository Structure - Milestone 2

13. Milestone 2 Completion Status

Deliverable	Status
CV parsing and structured extraction (LLM + regex)	✅Complete
Folder-based CV ingestion with batch processing	✅Complete
Educational profile analysis - normalization, gaps, institution quality	✅Complete
Professional experience analysis - timeline, gaps, overlaps, progression	✅Complete
Missing information detection - three-tier classification	✅Complete
Personalized email drafting per candidate	✅Complete
Initial candidate summary generation with suitability label	✅Complete
Partial research profile processing	✅Complete
Tabular outputs - education, experience, publications	✅Complete
Initial charts - score charts, publication timeline, radar chart	✅Complete
Intermediate web application - 7-tab dashboard	✅Complete
Advanced research analysis (WoS, Scopus, co-authorship)	□ Milestone 3
Full graphical dashboard and candidate ranking	□ Milestone 3

Table 5: Milestone 2 Completion Checklist

14. Limitations and Future Work

14.1 Current Limitations

- Institution quality lookup uses a curated static database. Universities not in the database return 'Not in database' - real-time QS/THE API integration is planned for Milestone 3.
- Publication title extraction from the NUST HR portal CVs occasionally includes co-author name fragments due to the multi-column table layout. LLM-based parsing reduces but does not eliminate this.
- Email and phone are not printed on portal-generated CVs - these are stored in the HR portal system and are correctly flagged as missing by the classifier.
- The Gemini API key must be configured for LLM-enhanced parsing. Without it, the system falls back to the regex parser automatically.
- Research profile analysis is partial - WoS/Scopus indexing, quartile ranking, and co-author network analysis are deferred to Milestone 3.

14.2 Planned for Milestone 3

- Full research profile analysis - journal WoS/Scopus indexing, quartile ranking (Q1–Q4), conference A* ranking via CORE portal, authorship role analysis
- Topic variability scoring and co-author network analysis
- Student supervision, books, and patents analysis
- Full graphical dashboard with comparative views across all candidates
- Candidate ranking module with weighted scoring across all dimensions
- Skill alignment analysis against job description
- Real-time institution quality via QS/THE API

15. Conclusion

Milestone 2 of TALASH has been successfully completed. The team has delivered a fully functional analysis pipeline covering educational profile analysis with CGPA normalization and institution quality assessment, professional experience analysis with timeline validation and gap detection, missing information detection with personalized email drafting, initial research profile processing, and candidate summary generation with suitability scoring.

The intermediate Streamlit web application provides a comprehensive 7-tab interface with batch CV processing, interactive Plotly charts, metric cards, a candidate selector, and a multi-candidate comparison dashboard with radar chart visualization. All Milestone 2 rubric criteria have been addressed, with the remaining advanced analysis modules - research profile depth, topic variability, and co-author analysis - planned for Milestone 3.

GitHub Repository: https://github.com/mehreennwhyn0t/Talash